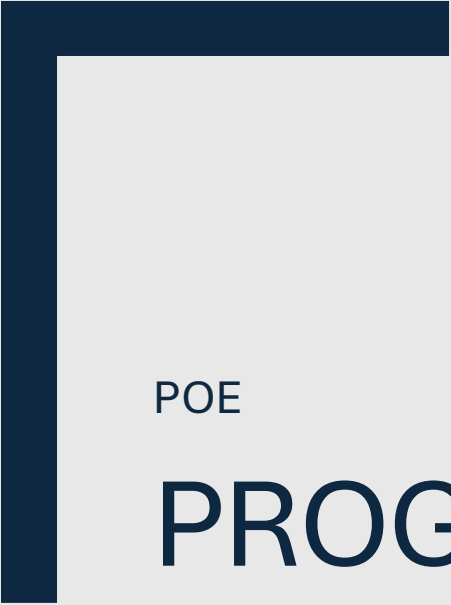




POE

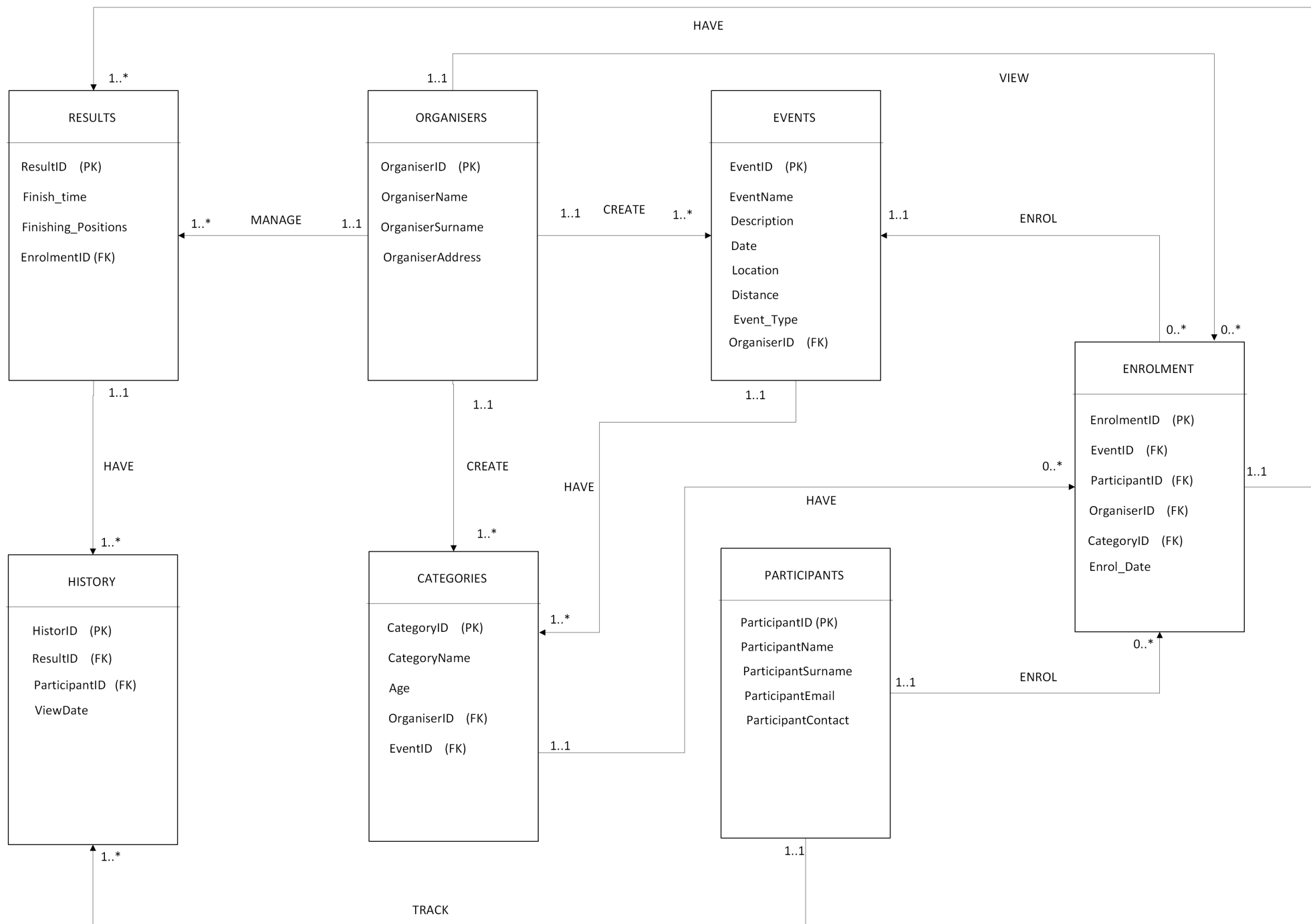
PROGRAMMING

Phetole Chalton Junior Kgatla

ST10476976 | GROUP 1



SECTION A- PART 1



SECTION B- PART 1

1 .PARTICIPANT ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/participants	Retrieves all participants.	Participant	None	200 OK – Returns list of participants.
GET	/api/participants/{id}	Retrieves a specific participant.	Participant	None	200 OK – Returns participant details. 404 Not Found – Participant does not exist.
POST	/api/participants	Creates a new participant.	Participant	{ "participantName", "participantSurname", "participantEmail", "participantContact" }	201 Created – Participant successfully created. 400 Bad Request – Invalid information. 409 Conflict – Participant already exists.
PUT	/api/participants/{id}	Updates participant information.	Organiser	{ "participantName", "participantSurname", "participantEmail", "participantContact" }	200 OK – Participant updated. 404 Not Found – Participant does not exist.
DELETE	/api/participants/{id}	Removes a participant.	Organiser	None	204 No Content – Participant deleted. 404 Not Found – Participant does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/participants/{id}/enrolments	Retrieves events entered by a participant.	Participant	None	200 OK – Returns participant's enrolments. 404 Not Found – Participant does not exist.
GET	/api/participants/{id}/history	Retrieves a participant's performance history.	Participant	None	200 OK – Returns performance history. 404 Not Found – Participant does not exist.

2. ORGANISERS ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/organisers	Retrieves all organisers.	Organiser	None	200 OK – Returns organisers.
GET	/api/organisers/{id}	Retrieves a specific organiser.	Organiser	None	200 OK – Returns organiser details. 404 Not Found – Organiser does not exist.
POST	/api/organisers	Creates an organiser.	Organiser	{ "organiserName", "organiserSurname", "organiserAddress" }	201 Created – Organiser created. 400 Bad Request – Invalid data.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/organisers/{id}	Updates organiser details.	Organiser	{ "organiserName", "organiserSurname", "organiserAddress" }	200 OK – Organiser updated. 404 Not Found – Organiser does not exist.
DELETE	/api/organisers/{id}	Deletes an organiser.	Organiser	None	204 No Content – Organiser deleted. 404 Not Found – Organiser does not exist.
GET	/api/organisers/{id}/events	Retrieves events created by an organiser.	Organiser	None	200 OK – Returns organiser's events. 404 Not Found – Event does not exist.
GET	/api/organisers/{id}/categories	Retrieves categories managed by an organiser.	Organiser	None	200 OK – Returns categories. 404 Not Found – Category does not exist.
GET	/api/organisers/{id}/results	Retrieves results managed by an organiser.	Organiser	None	200 OK – Returns results. 404 Not Found – Results does not exist.

3.Events Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves all available events.	None	None	200 OK – Returns list of events.
GET	/api/events/{id}	Retrieves details of a specific event.	None	None	200 OK – Returns event. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new event.	Organiser	{ "eventName", "description", "date", "location", "distance", "eventType", "organiserId" }	201 Created – Event created. 400 Bad Request – Invalid data.
PUT	/api/events/{id}	Updates an existing event.	Organiser	{ "eventName", "description", "date", "location", "distance", "eventType" }	200 OK – Event updated. 404 Not Found – Event does not exist.
DELETE	/api/events/{id}	Deletes an event.	Organiser	None	204 No Content – Event deleted. 404 Not Found – Event does not exist.
GET	/api/events/{id}/categories	Retrieves categories belonging to an event.	None	None	200 OK – Returns categories. 404 Not

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/enrolments	Retrieves participants enrolled in an event.	Organiser	None	Found – Event does not exist. 200 OK – Returns enrolments. 404 Not Found – Event does not exist.
GET	/api/events/{id}/results	Retrieves results for an event.	None	None	200 OK – Returns event results. 404 Not Found – Event does not exist.

4.CATEGORIES ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Retrieves all categories.	None	None	200 OK – Returns categories.
GET	/api/categories/{id}	Retrieves a specific category.	None	None	200 OK – Returns category. 404 Not Found – Category does not exist.
POST	/api/categories	Creates a category for an event.	Organiser	{ "categoryName", "age", "organiserId", "eventId" }	201 Created – Category created. 400 Bad Request – Invalid data. 404 Not Found – Event/organiser does not exist.
PUT	/api/categories/{id}	Updates a category.	Organiser	{ "categoryName", "age" }	200 OK – Category updated. 404 Not Found – Category does not exist.
DELETE	/api/categories/{id}	Deletes a category.	Organiser	None	204 No Content – Category deleted. 404 Not Found – Category does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					Found – Category does not exist.

5. ENROLMENT ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments	Retrieves enrolments.	Organiser	None	200 OK – Returns enrolments.
GET	/api/enrolments/{id}	Retrieves a specific enrolment.	Participant/Organiser	None	200 OK – Returns enrolment. 404 Not Found – Enrolment does not exist.
POST	/api/enrolments	Enrols a participant into an event and category.	Participant	{ "eventId", "participantId", "organiserId", "categoryId", "enrolDate" }	201 Created – Enrolment created. 404 Not Found – Event/participant/category does not exist. 409 Conflict – Participant already enrolled.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/enrolments/{id}	Updates an enrolment.	Participant	{ "categoryId" }	200 OK – Enrolment updated. 404 Not Found – Enrolment does not exist.
DELETE	/api/enrolments/{id}	Cancels an enrolment.	Participant	None	204 No Content – Enrolment cancelled. 404 Not Found – Enrolment does not exist.

6.RESULTS ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/results	Retrieves all race results.	None	None	200 OK – Returns results.
GET	/api/results/{id}	Retrieves a specific result.	None	None	200 OK – Returns result. 404 Not Found – Result does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Records a participant's result.	Organiser	{ "finishTime", "finishingPosition", "enrolmentId" }	201 Created – Result recorded. 404 Not Found – Enrolment does not exist.
PUT	/api/results/{id}	Updates a participant's result.	Organiser	{ "finishTime", "finishingPosition" }	200 OK – Result updated. 404 Not Found – Result does not exist.
DELETE	/api/results/{id}	Deletes a result.	Organiser	None	204 No Content – Result deleted. 404 Not Found – Result does not exist.
GET	/api/enrolments/{id}/result	Retrieves the result for an enrolment.	Participant/Organiser	None	200 OK – Returns result. 404 Not Found – Result does not exist.

7. HISTORY ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/history/{id}	Retrieves a history record.	Participant	None	200 OK – Returns history record. 404 Not Found – History does not exist.
GET	/api/participants/{participantId}/history	Retrieves a participant's performance history.	Participant	None	200 OK – Returns history. 404 Not Found – History does not exist.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					Found – Participant does not exist.

SECTION C- PART 1

```
--CREATE DATABASE RaceDay;
```

```
--CREATE TABLE ORGANISERS(
--OrganiserID INT NOT NULL PRIMARY KEY,
--OrganiserName VARCHAR(30) NOT NULL,
--OrganiserSurname VARCHAR(30) NOT NULL,
--OrganiserAddress VARCHAR(60) NOT NULL
--);
```

```
--CREATE TABLE PARTICIPANTS(
--ParticipantID INT NOT NULL PRIMARY KEY,
--ParticipantName VARCHAR(30) NOT NULL,
--ParticipantSurname VARCHAR(30) NOT NULL,
--ParticipantEmail VARCHAR(30),
--ParticipantContact INT NOT NULL
--);
```

```
--CREATE TABLE EVENTS(
--EventID INT NOT NULL PRIMARY KEY,
--EventName VARCHAR(80) NOT NULL,
--[Description] VARCHAR(100) NOT NULL,
--[Date] VARCHAR(9) NOT NULL,
--[Location] VARCHAR(40) NOT NULL,
--Distance VARCHAR(5000) NOT NULL,
--EventType VARCHAR(30),
--OrganiserID INT NOT NULL,
```

```
--FOREIGN KEY (OrganiserID) REFERENCES ORGANISERS(OrganiserID)
--);
```

```
--CREATE TABLE CATEGORIES(
--CategoryID INT NOT NULL PRIMARY KEY,
--CategoryName VARCHAR(30) NOT NULL,
--Age INT NOT NULL,
--OrganiserID INT NOT NULL,
--EventID INT NOT NULL,
--SELECT * FROM CATEGORIES;
```

```

--FOREIGN KEY (OrganiserID) REFERENCES ORGANISERS(OrganiserID),
--FOREIGN KEY (EventID) REFERENCES EVENTS(EventID)
--);

--CREATE TABLE ENROLMENT(
--EnrolmentID INT NOT NULL PRIMARY KEY,
--EventID INT NOT NULL,
--ParticipantID INT NOT NULL,
--OrganiserID INT NOT NULL,
--CategoryID INT NOT NULL,
--EnrolDate VARCHAR(9) NOT NULL,

--FOREIGN KEY (EventID) REFERENCES EVENTS(EventID),
--FOREIGN KEY (ParticipantID) REFERENCES PARTICIPANTS(ParticipantID),
--FOREIGN KEY (OrganiserID) REFERENCES ORGANISERS(OrganiserID),
--FOREIGN KEY (CategoryID) REFERENCES CATEGORIES(CategoryID)
--);

--CREATE TABLE RESULTS(
--ResultID INT NOT NULL PRIMARY KEY,
--Finish_time VARCHAR(20) NOT NULL,
--Finishing_Positions INT NOT NULL,
--EnrolmentID INT NOT NULL,

--FOREIGN KEY (EnrolmentID) REFERENCES ENROLMENT(EnrolmentID)
--);

--CREATE TABLE HISTORY(
--HistoryID INT NOT NULL PRIMARY KEY,
--ResultID INT NOT NULL,
--ParticipantID INT NOT NULL,
--ViewDate VARCHAR(9) NOT NULL,

--FOREIGN KEY (ResultID) REFERENCES RESULTS(ResultID),
--FOREIGN KEY (ParticipantID) REFERENCES PARTICIPANTS(ParticipantID)
--);

--INSERT INTO ORGANISERS
--(OrganiserID,OrganiserName,OrganiserSurname,OrganiserAddress)
--VALUES
--(1,'Junior','Kgatla','4972 Mathibestad'),
--(2,'Orapeleng','Rakgekola','4973 Mathibestad');
--SELECT * FROM ORGANISERS;

--INSERT INTO PARTICIPANTS
--(ParticipantID,ParticipantName,ParticipantSurname,ParticipantEmail,ParticipantContact)
--VALUES
--(1,'Chalton','Modika','CharlesModika@gmail.com',0768832210),
--(2,'Charllot','lesenya','lesenya@gmail.com',0768543210);
--SELECT * FROM PARTICIPANTS;

--INSERT INTO EVENTS

```

```
--(EventID,EventName,[Description],[Date],[Location],Distance,EventType,OrganiserID)
--VALUES
--(1,'Pretoria City Run','20KM Run','03/10/26','Pretoria CBD','20KM','Running',2),
--(2,'Hammanskraal Fun Run','5KM Fun Run','15/10/26','Hammanskraal Sports Ground','5KM','Fun Run',1),
--(3,'Tshwane Marathon','15KM Marathon','23/10/26','Soshanghuve Giant Stadium','15KM','Marathon',2);
--(10001,'Pretoria City Run','20KM Run','03/10/26','Pretoria CBD','20KM','Running',2),
--(10002,'Hammanskraal Fun Run','5KM Fun Run','15/10/26','Hammanskraal Sports Ground','5KM','Fun Run',1),
--(10003,'Tshwane Marathon','15KM Marathon','23/10/26','Soshanghuve Giant Stadium','15KM','Marathon',2);
--SELECT * FROM EVENTS;

--INSERT INTO CATEGORIES
--(CategoryID, CategoryName,OrganiserID,EventID,Age)
--VALUES
--(1,'Senior',2,1,'18+'),
--(2,'Junior',1,2,'Under 20'),
--(3,'Senior',2,3,'Above 18');
--SELECT * FROM CATEGORIES;

--INSERT INTO ENROLMENT
--(EnrolmentID,EventID,ParticipantID,OrganiserID,CategoryID,EnrolDate)
--VALUES
--(1,3,1,2,3,'10/10/26');
--SELECT * FROM ENROLMENT;
```